THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. How many seconds are there in 24 hours? 1

(a) 86400 **(b)** 86460 **(c)** 90000 **(d)** 86340

2. The expression whose value is 10 is: 1

(a) $(20 - 5) \times 2$ **(b)** $20 - 5 \times 2$ **(c)** $20 \times 2 - 5$ **(d)** $20 - 5 + 2$

3. 3 m 45 cm expressed in m is: 1

(a) 3.35 **(b)** 3.45 **(c)** 4.45 **(d)** 3.55

4. A pencil costs ₹x and an eraser costs ₹y. The cost of 3 pencils and 5 erasers is: 1

(a) $15(x + y)$ **(b)** $8xy$ **(c)** $3y + 5x$ **(d)** $3x + 5y$

5. A transversal makes alternate interior angles of 65° and 65°. The two lines are: 1

(a) perpendicular **(b)** not parallel **(c)** parallel **(d)** intersecting

6. For every whole number n, $n^2 + n$ is always: 1

(a) even **(b)** a prime number **(c)** a multiple of 3 **(d)** odd

7. In two triangles, $AB = PQ$, $BC = QR$, $\angle B = \angle Q$. They are congruent by: 1

(a) not congruent **(b)** ASA **(c)** SSS **(d)** SAS

8. Which is the greatest? 1

(a) $\frac{3}{4} \times \frac{1}{8}$ **(b)** $\frac{1}{8}$ **(c)** $\frac{3}{4} \div \frac{1}{8}$ **(d)** $\frac{3}{4}$

9. The number of days in 10 years (non-leap years of 365 days) is: 1

(a) 3650 **(b)** 3285 **(c)** 4015 **(d)** 3750

10. The value of $(84 - 12) \div (3 \times 4) + 9$ is: 1

(a) 15 **(b)** 77 **(c)** 17 **(d)** 105

11. 3.456 rounded to two decimal places is: 1

(a) 3.56 **(b)** 3.46 **(c)** 4.46 **(d)** 3.36

12. If $a = 3$, the value of $a^2 + 2a$ is: 1

(a) 15 **(b)** 17 **(c)** 14 **(d)** 16

13. Two supplementary angles differ by 20°. The larger angle is: 1

(a) 90° **(b)** 120° **(c)** 110° **(d)** 100°

14. A number has digit sum 12. It is divisible by: 1

(a) 9 but not 3 **(b)** both 3 and 9 **(c)** neither 3 nor 9 **(d)** 3 but not 9

15. An isosceles triangle has equal sides 9 cm and base 6 cm. Its perimeter (cm) is: 1

(a) 26 **(b)** 23 **(c)** 24 **(d)** 25

16. A car travels $\frac{35}{2}$ km on 1 litre of petrol. On 6 litres it travels (km): 1

(a) 105 **(b)** 210 **(c)** $52\frac{1}{2}$ **(d)** $105\frac{1}{2}$

17. 2,30,000 × 15 = 1

(a) 34,51,000 **(b)** 3,45,00,000 **(c)** 34,50,000 **(d)** 34,40,000

18. The value of $5 \times (18 - (3 + 7)) - 12$ is: 1

(a) 28 **(b)** 26 **(c)** 82 **(d)** 30

19. Which number lies between 0.3 and 0.4? 1

(a) 0.37 **(b)** 0.43 **(c)** 0.3 **(d)** 0.04

20. If $x = 5$, the value of $\frac{4x+2}{2}$ is: 1

(a) 13 **(b)** 10 **(c)** 12 **(d)** 11

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Two angles forming a linear pair differ by $20°$. Find both angles. 2

22. The sum of three consecutive odd numbers is 51. Find them. 2

23. The exterior angle of a triangle is $(4x - 10)°$ and the interior opposite angles are $(x + 12)°$ and $(2x + 18)°$. Find x and the exterior angle. 2

24. $\frac{3}{4}$ of a number is 27. Find the number and also $\frac{1}{4}$ of it. 2

25. Find the number of minutes in 30 days. Write it in the Indian system. 2

26. Check whether $12 \times (9 - 4)$ and $12 \times 9 - 12 \times 4$ are equal. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Find 34,56,789 + 23,45,678 - 12,34,567. Estimate the answer by rounding each number to the nearest lakh and compare it with the exact answer. 3

28. Insert one pair of brackets in $2 + 3 \times 4 + 5$ so that its value is 25. 3

29. Arrange $4.05, 4.5, 4.005, 4.55$ in descending order and find the difference between the greatest and the smallest. 3

30. A cab charges ₹45 plus ₹11 per km. Write the fare for k km. Find the fare for 8 km and for 13 km. 3

31. Two supplementary angles are in the ratio $1 : 3$. Find both angles and the complement of the smaller one. 3

OR

Write 75 as the sum of 5 consecutive numbers. 3

32. $\triangle ABC \cong \triangle DEF$ with $AB = (5x + 3)$ cm, $DE = (2x + 18)$ cm, $BC = 7$ cm and $AC = 9$ cm. Find x, the length DE and the perimeter of $\triangle DEF$. 3

OR

Evaluate $\left(\frac{1}{2}+\frac{1}{3}\right)\times\frac{6}{5}$. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. A wire 7.8 m long is cut into pieces of 0.9 m each. How many whole pieces are made and what length is left over? 5

34. A shopkeeper sells x kg of rice at ₹45 per kg and y kg of wheat at ₹30 per kg. Write an expression for his total sales and find it when $x = 20$ and $y = 10$. 5

35. Two interior angles on the same side of a transversal cutting two parallel lines are in the ratio $1:5$. Find both angles and the angle forming a linear pair with the larger one. 5

36. Write the Collatz sequence from 20. Find the largest number reached, the number of even terms and the number of odd terms. 5

OR

The equal sides of an isosceles triangle are $(2x+2)$ cm and $(3x-3)$ cm and its base is $(x+6)$ cm. Find x, the three sides and the perimeter. Check that the triangle exists. 5

37. A wall of 180 m^2 is painted in three days. On day 1, $\frac{2}{5}$ of it is painted; on day 2, $\frac{1}{3}$ of the remainder; the rest on day 3. Find the area painted on each day. 5

OR

The distance between two cities is 24,50,000 m. A runner covers 65,000 m each day. How many full days does he run, how much distance is left after that, and how many days in all are needed to finish? 5

38. The expression $(500-3\times60)\div4$ models a situation. Evaluate it, and then write a story problem for it (one line) and give the answer with units. 5

OR

A tank of capacity 72 litres is filled by a tap at 4.8 litres per minute. How many minutes does it take to fill the empty tank? How much water is in it after 9 minutes and how much more is needed? 5

END OF QUESTION PAPER Total: 80 Marks

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